

Test Automation Strategy 2025-2026

 Note to reader:

This document presents version 1.0 of the Test Automation Strategy for the years 2025-2026. It is important to note that this document will undergo light revisions in the coming year. The next update is scheduled for Q2, following the evaluation of the test automation maturity survey results.

1. Purpose

The purpose of this document is to ensure that test automation in LocalTapiola is implemented effectively and efficiently within a software development lifecycle. It provides clear plan and framework to guide teams in automating repetitive testing, improving the speed and the accuracy of testing, and thus supporting continuous integration and delivery.

 High-level Vision

The standardization of test automation at LocalTapiola is set to enhance both the speed and reliability of our testing processes, ultimately elevating the quality of the systems developed for our customers. By implementing a unified testing framework and infrastructure, we are positioning ourselves to support release on-demand model and achieving true shift-left practices, paving the way for cost-effective test automation in the future.

The success of test automation development hinges on excellent collaboration between LocalTapiola and our vendors. This partnership is crucial for fostering a cultural shift that positions test automation as a cornerstone of our quality assurance practices. Furthermore, we are actively exploring the potential of artificial intelligence to supercharge our test automation efforts, ensuring that we remain on the right path moving forward.

Presentations

File	Modified
 test_automation_strategy_cornerstones_v1_20122024.pptx	Dec 20, 2024 by Paavo Rohamo

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2. Scope

The current scope of this test automation strategy document includes functional testing types like smoke, system, integration, and user acceptance testing. However, it does not address unit, security, accessibility, load, or performance testing, which may be automated at some point in the future.

Testing levels included:

Smoke

System

Integration

User Acceptance

3. Objectives

The primary aim of this test automation strategy is to create cohesive standards, methodologies, frameworks, and platforms for test automation at LocalTapiola. Additionally, it will steer various stakeholders involved in test automation towards LocalTapiola's direction. We have pinpointed shared test automation goals and outlined four key objectives that will form the cornerstone of this document.

Shared objectives for test automation

- Increase test coverage
 - Automate repetitive test cases and scenarios and focus on high-risk and high-value areas of the applications
 - Ensure that those parts of the application which are not possible to be tested via manual testing are automated
- Speed up testing cycles, especially regression testing
 - Automate time-consuming and repetitive manual tests
- Early defect detection
 - Shift-left, more testing at system and integration testing levels.
- Faster regression testing and feedback
 - Test cases and sets must be robust, parallelized, and scalable.
- Optimize resource utilization
- Reduce costs over time
 - Automate repetitive test cases
 - Create maintainable and scalable automations

1. Standardization and practices

- **Unified test automation standards, models and processes**
 - Creating unified processes and models for everyone doing test automation practices
 - Improved quality, scalability, maintainability
- **Standardised frameworks, tools and platforms**
 - **Robot Framework** should be used as the **main test automation framework** in the future at LocalTapiola
 - Unified testing framework will allow true shift left and test scripts produced in earlier testing phases can be used in later stages.
 - **Secondary test automation framework** is **Java+Selenium framework**, but the goal will be just one unified testing framework in the future

- Potential migrations out from Java+Selenium framework has to be carefully planned and executed. Those will require great efforts from all parties included. Migrations has to be really carefully evaluated and has to have real benefits for LocalTapiola, and should not bring additional costs to projects.
- Seek and evaluate potential future low-code test automation frameworks with builtin AI functionalities
- **Azure DevOps Pipelines** and **Azure DevOps Git** should be the main tools for CI/CD and Version Management
- **Custom Robot Framework libraries** for common testing functionalities
 - Usage of community made libraries is encouraged
 - Creation of LocalTapiolas own Robot Framework libraries for common components (Core system navigations, logins, etc.)
 - •Already under development (CustomerLibrary, ZephyrLibrary)
- **Common documentations** for tools, infra and usage should be created
 - Test automation results must be presented transparently for those in need
 - Running test automations and reviewing results must happen in hours, not days
 - Goal for pipeline run is (1h at max)
 - Migrating to AzureDevOps pipelines environment

2. Collaboration and ecosystem

• **Test automation ownership and governance**

- LocalTapiola will be deepening vendor guidance while unifying test automation framework. LocalTapiola will make test automation guidelines for processes, models and tools.
- We must ensure better collaboration between LT test automation and vendors, and also between vendors. This enables knowledge sharing between all test automation related parties and business.
- Getting rid of silos that currently exists.
 - Current silos affect greatly on the price tag of test automation

3. Competence and test automation culture

- Creating test automation culture is long term and continuous practice
 - True **Shift Left** enabled by unified test automation framework
 - Leading by example
 - LocalTapiolas own team will do its best to champion, adapt and share test automation culture at LocalTapiola

- We are open to all kinds of ideas and suggestions from our vendors regarding culture and competence development
- Journey for test automation culture requires
 - Standardizing test automation framework, processes and models
 - Enabling faster test automation runs and results
 - Integrating automation into everyday practices → Dashboards and health metrics
 - Competence
 - Training sessions for LocalTapiola Quality Assurance
 - Enabling professional path for manual testers
 - Requires continuous development
- **Feedback loop** for test automation
 - **Test automation maturity survey**
 - Ensuring time and budget for refactoring and technical debt
 - This requires discussions with business

4. Test Data Management and Artificial Intelligence

- **Test data used should be synthetic**
 - Automations should create and delete their own test data used
 - Goal is that no masked test data is used in the future
- **Upcoming projects**
 - Creation of test data with automation tools (2025 Q1 →)
 - Using AI as part of test automation maintenance (2025 Q1-Q2 →)
 - Hopefully we get one dedicated LLM which could be used by all test automation related matters
- **Research**
 - Using AI in test case and test code generation
 - Possible future test automation frameworks utilizing AI
 - Researching future Robot Framework libraries that uses AI at some level

4. Testing Levels

 Testing levels are covering only functional testing.

Non-functional testing is out of scope for this test automation plan.

Unit testing or automated unit testing is not in the scope of this test automation strategy.

	Smoke	System- Integration (SIT)	Integration	User Acceptance (UAT)
Purpose	Check if the basic functionality of the application is working.	Test end-to-end functionalities of the applications. Validates the entire systems functionality against requirements.	Test integrations between integrated modules such as API's, ensures different modules and components work together as intended.	Validates the system against business requirements and end-user expectations
Automation focus	Quick high-level checks: • APIs are responding • Application / system launches and display login or main page	Automated functional testing of workflows and e2e scenarios. Focus on critical paths and high-risk areas first	• APIs • Middleware • Database interactions	Automating regression tests to ensure new changes don't break existing functionality At this level we should not check every check box and form field is functional, but that the e2e flows between different business systems are functional Focus on critical paths and high-risk areas first

Benefits	• Immediate feedback after build • Avoids wasted effort on deeper testing if basic functionality fails (in Azure DevOps platform this is especially important cost wise)	Reduces manual effort for large-scale system checks. Ensures consistency and enables regression testing for continuous development.	Able to verify complex interactions are working as intended. Ensures stability in integrations.	Using test automation at regression testing enables faster production times and more stable systems.
Responsibility	• LocalTapiola, Vendors Before commencing full System or User Acceptance test runs, it is essential to conduct smoke tests to ensure that the basic functionalities are working as intended.	• Vendors The primary responsibility for system testing lies with the vendors.	• Vendors, (LocalTapiola) The primary responsibility for system testing lies with the vendors. LT test automation has to make some integration testing to be able to create sufficient UAT regression test sets.	• LocalTapiola, (Vendors)
Practices	Incorporate a smoke test phase into your pipelines and execute it prior to running the test sets. This practice will help verify that the test environment is stable, reducing	At this level we should verify functionality of components like: • Forms, Text Fields, Inputs, Buttons • Navigation components	Use industry best practices. Create “smoke” tests for APIs and other systems If tool used is Robot Framework: • Prefer templates	To ensure that run times remain efficient and manageable, it's essential to create more targeted and focused test cases. To ensure that business functionalities

	the likelihood of failures in the subsequent test sets.	Invalid inputs, edge cases, error conditions Encapsulate UI interactions in reusable components (Page Object Model, or similar) To ensure that run times remain efficient and manageable, it's essential to create more targeted and focused test cases.	- Do not hardcode, paths, keys, headers, etc. - Use modular code with ability to scale	operate efficiently, it is essential to maximize the utilization of APIs. This approach will also contribute to maintaining low test run times.
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5. Test Environments

Test automation should be thoughtfully designed and implemented to enhance the usability and scalability of test suites and cases across different environments. We should be able to use the same test automation test suites and cases in every possible environment with slight changes. To achieve this, it is essential to store environment variables/urls/paths in variables rather than hard-coding them directly into test cases.

Test environments used in LocalTapiola for test automation are:

- SIT
- SIT2
- UAT

6. Test Case Design

Best Practices and standardization

- Test case design should follow the best practices and industry standards for given test automation framework.
- Focus on critical paths and high-risk areas first

- To ensure optimal performance, each test case should be designed to run independently. This approach facilitates complete parallelization, allowing tests to execute without dependencies on previous cases.
- Use proper test and suite; setup and teardown methods
- To effectively manage dynamic elements and timing challenges, it is advisable to utilize explicit waits and "wait until" conditions. Avoid relying on sleep functions whenever possible.

Maintainability

- Test cases should be maintainable, scalable and modular.
- Ensure test scripts are easy to read, understand, documented at sufficient level.
- Our objective is to ensure that it doesn't matter who created a test case; with standardisation and usage of best practices every test automation developer at LocalTapiola should be equipped with the knowledge to resolve any specific test case
- As we move towards a unified test automation framework, it is essential to prioritize design for reusability. Our objective is to leverage the test automation scripts created during the system and integration testing phases for use in the User Acceptance Testing (UAT) phase.
- Separate test data from test logic to enable reusability and maintainability

Robot Framework best practices

Robot Framework employs a "Keyword Driven" approach for test automation scripting, which should be consistently applied at every level within LocalTapiola.

For best practices related to Robot Framework, you can refer to the [Style Guide | ROBOT FRAMEWORK](#) and the [Robot Framework User Guide](#).

Additionally, you can consult our own Robot Framework style guide for further insights.

Robot Framework style guide

LT has created style guide and project template which should be used as a base when test automation is conducted through Robot Framework. These documents will be under constant development.

- Style Guide:
 - Confluence: [Robot Framework Style Guide - WIP - Testing - Confluence \(atlassian.net\)](#)
 - GIT: [style_guide - Repos \(visualstudio.com\)](#), if you need access message owner of this document.
- Project Template:
 - GIT: [style_guide - Repos \(visualstudio.com\)](#), if you need access message owner of this document.

7. Test Data Management

Test automation should, by default, generate the necessary test data used. This approach aims to ensure that all test data at LocalTapiola is synthetic in the future. However in some cases generation of test data is not possible for more complex cases. These situations has to be evaluated carefully, as the goal is that no masked data is used in the future.

Both LocalTapiola and its vendors should work collaboratively towards establishing standardized methods for test data creation.

NOTE: This section will be updated after H1/2025 Test Data project.

8. Test Automation Tools

Robot Framework (Primary test automation framework)

Robot Framework is set to become the primary test automation framework for the foreseeable future.

Currently, many system-level test automations are built using the Java + Selenium test automation framework. It is essential to explore the most effective methods for maintaining Java + Selenium at an optimal level while also investigating the best approaches to transition these tests to Robot Framework. However, it is important to note that in some instances, such a conversion may not be feasible.

Java + Selenium testing frameworks (Secondary test automation framework)

Java and Selenium testing frameworks are primarily employed by TCS and Toivo projects. LocalTapiola continues to utilize Java and Selenium in ongoing projects, as well as for the regression sets developed in the past, as long as necessary.

LocalTapiola is set to begin planning for potential migrations from Java + Selenium to the Robot Framework in the Spring of 2025. The initial projects under consideration for this migration include Aito Digi, other Digi projects, and Toivo. At this stage, we lack clear visibility on whether all projects can be migrated and the specific timelines for each transition.

It is recommended that no new test automation projects utilizing Java and Selenium be initiated.

Browserstack

BrowserStack offers a comprehensive solution for testing across a diverse range of browsers and mobile devices.

While the use of BrowserStack is highly encouraged, it is not a strict requirement. LocalTapiola is committed to developing templates for the Robot Framework, which facilitate seamless integration with BrowserStack whenever necessary.

Azure DevOps pipelines

LocalTapiola has made substantial investments in migrating to a cloud platform. Currently, many of our regression test automations are executed on local servers. Our objective is to transition from these servers to utilizing Azure DevOps environments.

This shift will significantly reduce the maintenance required for various environments, such as Jenkins.

Transitioning to Azure DevOps pipelines presents considerable advantages compared to other tools. By moving away from Jenkins pipelines, we can not only improve the scalability of our DevOps practices across different projects but also lower long-term infrastructure maintenance costs.

Azure DevOps pipelines enable DevOps teams at LocalTapiola to implement a release-on-demand model. As a result, migrations will be pivotal in this transition.

Azure Keyvault

Utilizing Azure Key Vault as the primary storage solution for various test automation-related secrets and passwords is highly recommended. Azure Key Vault can be seamlessly integrated into Azure DevOps pipelines, allowing for the retrieval of secrets without the need for managing additional secret files. This approach simplifies the process, especially when passwords and secrets require updates.

Utilizing tools such as Keepass raises significant security concerns, primarily due to the challenges in managing access to passwords, sensitive information, and user rights.

LocalTapiola strongly advocates for transitioning away from tools like Keepass as swiftly as possible.

9. Documentation

Test automation must be documented properly, adhering to established best practices. It should be regarded as a form of software development rather than merely a collection of scripts or gimmicks. Proper documentation will significantly ease maintenance efforts. This chapter outlines key principles to follow in LocalTapiola's test automation, which can certainly be expanded upon.

Tool / Framework	Documentation
Test case	Jira / Zephyr
Robot Framework	Test cases according to Robot Framework style guide (test case should document itself) Libraries according to python PEP8 standards Commenting code (make comments when necessary, do not overuse comments)
Project test automation LifeCycle	Project confluence pages
UAT test automation LifeCycle documentation	See this page: Projects UNDEFINED

10. Reporting & Metrics

a. Test automation reporting

Zephyr

Zephyr is used as the primary channel for reporting test automation results to stakeholders.

Currently, the guidelines for using Zephyr vary among users. Our goal is to standardize these methods while aligning our test automation framework with the Robot Framework.

Other reporting methods

As we increasingly engage with Azure DevOps pipelines, it is essential to explore opportunities for enhancing the visibility of our reporting to stakeholders and the business. Additionally, we require clear dashboards and health metrics to serve our internal needs effectively.

Exploring alternative methods to enhance test automation reporting is highly encouraged. By doing so, you can achieve greater robustness, speed, and transparency in your reporting processes.

b. Test automation maturity metric

Introducing a new metric for assessing test automation maturity. Tracking these KPIs helps LocalTapiola to understand our test automation maturity, identify areas for improvement and demonstrate the value of automation over time. This will also help us to keep track of future improvements made in the field of test automation at LocalTapiola.



- **Why:** To gain insights of test automation maturity at LocalTapiola and to assess its current state.
- **When:** Twice a year, at the beginning of each half.
 - Dates:
 - 2025 H1
 - 1st Question form will be sent on 27.01.2025 (time to answer 1 week)
 - Going through the results during Q1, and making improvements to strategy
 - Results Presentation: Week 9
 - 2nd Question form will be sent on 01.08.2025 (time to answer 2 weeks)
 - Going through the results during Q3, and making improvements to strategy
 - Results Presentation: Week 37-38
- **How:** Inquiries targeting test automation developers and test managers will be conducted using a form based on test automation maturity KPIs. The completed forms will be sent via email, and the results will be thoroughly evaluated.
- **By who:** The head of test automation at LocalTapiola will initiate inquiries, and the results will be evaluated in partnership with the Quality Assurance managers.
- **Purpose:**
 - *Elevating Test Automation Practices to New Heights: Gaining Strategic Insights into Existing Gaps and Areas for Improvement*

Test Automation Maturity KPI's

- **SQ1-Strategy**
 - We have a test automation strategy that defines 'what test scope will be automated to what degree, when, by whom, by which methods, by what test tools, in what kind of environment'.
- **SQ2-Resources**
 - We allocate enough resources for test automation, e.g., skilled people, the funding, the time & effort, test environment with the required software, hardware, or test data for test automation.

- **SQ3-Roles**

- We clearly define roles and responsibilities of stakeholders in test automation.

- **SQ4-Knowledge**

- We are systematically learning from prior projects. We collect and share expertise, good test automation practices, and good test tools for future projects.

- **SQ5-Competence**

- Our test team has enough expertise and technical skills to build test automation based on our requirements.

- **SQ6-Tools**

- We currently have the right test tools that best suit our needs.

- **SQ7-Test environment**

- We have control over the configuration of our test environment.

- **SQ8-Guidelines**

- We have guidelines on designing and executing automated tests. Those guidelines include, e.g., coding standards, test-data handling methods, specific test design techniques to create test cases, processes for reporting and storing test results, the general rules for test tool usage, or information on how to access external resources.

- **SQ9-Prioritization**

- We effectively prioritize and schedule automated tests for the execution.

- **SQ10-Results**

- We are capable to manage and integrate test results collected from different sources (e.g., different test tools, test levels, test phases) into a big picture, and then report useful information to the relevant stakeholders.

- **SQ11-Process**

- We organize our test automation activities in the stable and controllable test process.

- **SQ12-SUT**

- Our Software Under Test enables us to conduct our test automation, e.g., maturity, running speed, or testability of our Software Under Test is not a problem for our test automation.

11. Training

From the perspective of a test automation developer, it is essential to offer training on the new libraries, tools, and methodologies developed for all stakeholders involved in test automation at LocalTapiola. This initiative will facilitate knowledge sharing and help dismantle existing silos within the team.

We are establishing a professional development pathway for manual testers at LocalTapiola. This initiative will enable our manual testers to cultivate a culture of test automation within the LocalTapiola Quality Assurance sector. Gaining even a minimal understanding of test automation practices can significantly enhance manual testers' comprehension of our automation efforts and objectives. This knowledge not only clarifies our processes but also provides valuable insights that can improve our testing practices.

The Head of Test Automation at LocalTapiola, along with the LocalTapiola Test Automation Team, will take charge of collecting ideas and resources for upcoming training sessions. We are always open to hearing ideas from our vendors regarding this matter and are eager to strengthen our culture through the training they provide.

12. Continuous Improvement

Continuous improvement in test automation necessitates a proactive strategy for identifying gaps, embracing innovation, and empowering teams. By cultivating a culture of enhancement and systematically tackling inefficiencies, LocalTapiola can transform our test automation into an invaluable asset for delivering high-quality software more efficiently.